Histograms

1 Construct a histogram to represent the following data sets:

a $3, 3, 4, 4, 4, 4, 5, 5, 5, 6, 6, 6, 6, 6, 7, 7, 8, 8, 8$

b $47, 45, 43, 47, 44, 43, 43, 45, 44, 47, 45, 42, 45, 44, 43, 44, 46, 44, 42, 46, 44, 45, 45, 42$

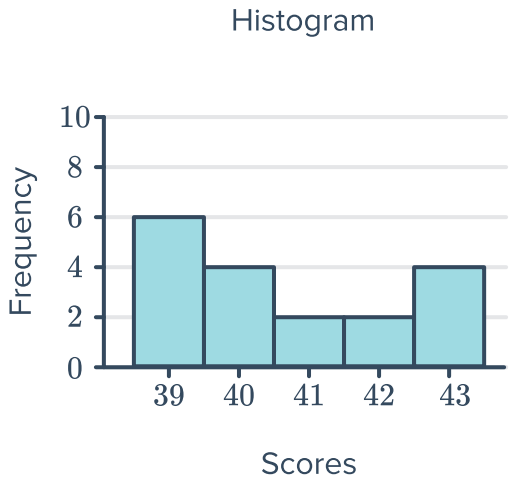
2 Consider the following histogram:

a State the frequency of the score of 43.

b Find the total number of scores.

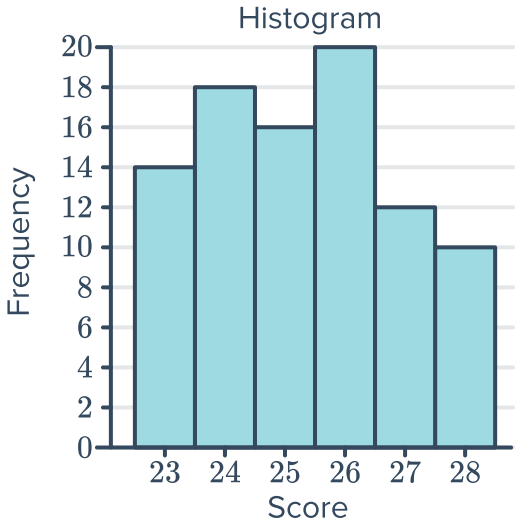
c Calculate the sum of the scores.

d Calculate the mean, correct to two decimal places.



3 Complete the frequency table for the data set in the histogram:

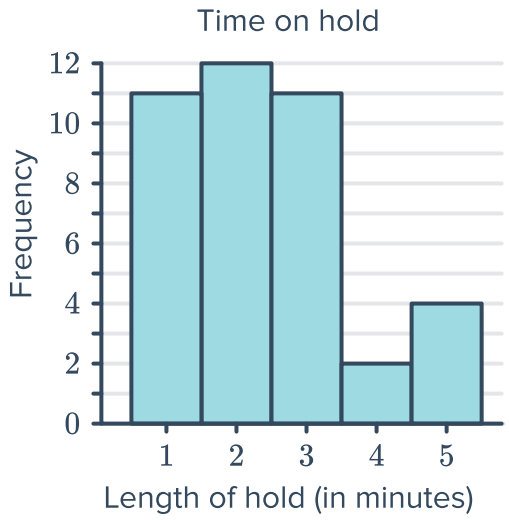
Score	Frequency
$22.5 \leq \text{Score} < 23.5$	
$23.5 \leq \text{Score} < 24.5$	
$24.5 \leq \text{Score} < 25.5$	
$25.5 \leq \text{Score} < 26.5$	
$26.5 \leq \text{Score} < 27.5$	
$27.5 \leq \text{Score} < 28.5$	



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4 A government agency records how long people wait on hold to speak to their representatives. The results are displayed in the given histogram:

Time on hold in minutes	Frequency
$0 \leq \text{Time} < 1.5$	
$1.5 \leq \text{Time} < 2.5$	
$2.5 \leq \text{Time} < 3.5$	
$3.5 \leq \text{Time} < 4.5$	
$4.5 \leq \text{Time} < 5.5$	



a Complete the frequency table based on the data from the histogram.

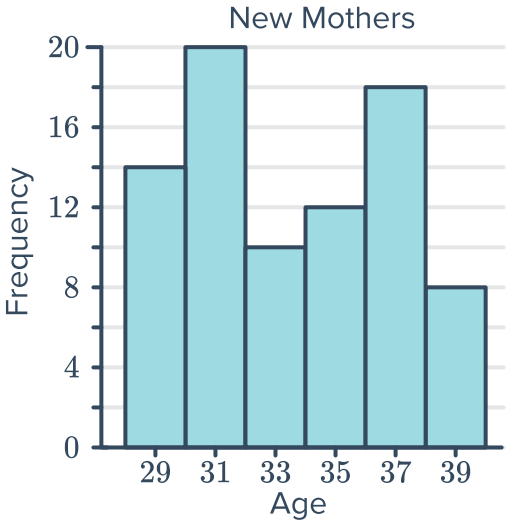
b How many phone calls were made?

c How many people were on hold for at least 2.5 minutes?

d How many people were on hold for less than 3.5 minutes?

5 The ages of new mothers in a particular hospital are displayed on the following histogram:

Age	Frequency
$28 \leq \text{Age} < 30$	
$30 \leq \text{Age} < 32$	
$32 \leq \text{Age} < 34$	
$34 \leq \text{Age} < 36$	
$36 \leq \text{Age} < 38$	
$38 \leq \text{Age} < 40$	



a Complete the frequency table based on the histogram.

b How many new mothers were there?

c How many mothers were under 34 years old?

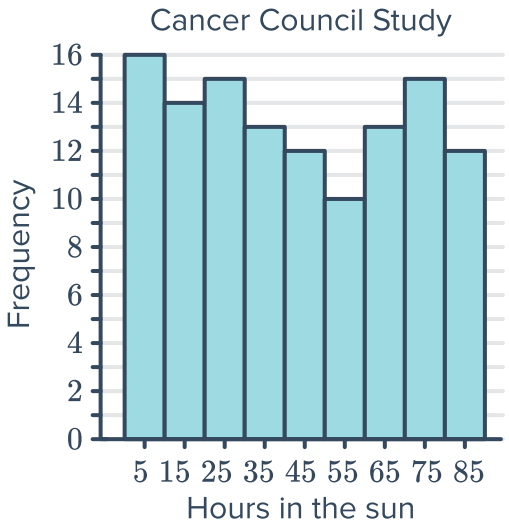
d How many mothers were 36 years old or older?

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6 The Cancer Council conducted a study to determine how long people spend in the sun in a month. They asked 120 participants to wear a detector at all times.

Their results are summarised in the given histogram:

Hours in Sun	Frequency
$0 \leq \text{Hours} < 10$	
$10 \leq \text{Hours} < 20$	
$20 \leq \text{Hours} < 30$	
$30 \leq \text{Hours} < 40$	
$40 \leq \text{Hours} < 50$	
$50 \leq \text{Hours} < 60$	
$60 \leq \text{Hours} < 70$	
$70 \leq \text{Hours} < 80$	
$80 \leq \text{Hours} < 90$	



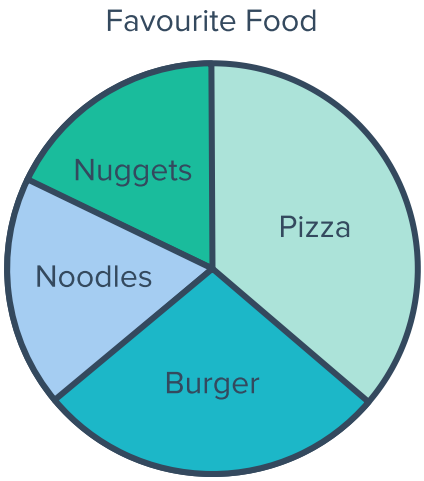
a Complete the frequency table based on the histogram.

b How many participants spent less than 30 hours in the sun? .

c What fraction of participants spent at least 60 hours in the sun? .

Pie charts

7 The given pie chart shows the results of a class survey where students were asked to nominate their favourite food:

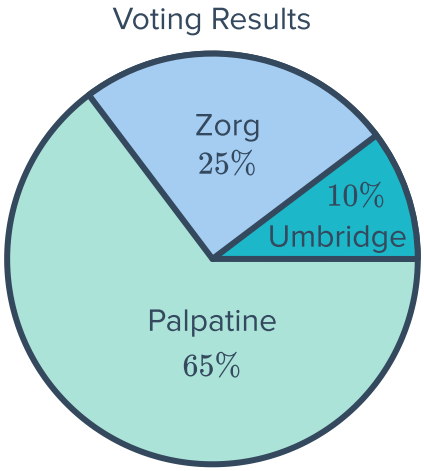


a Which was the most popular food?

b Which two foods were equally popular?

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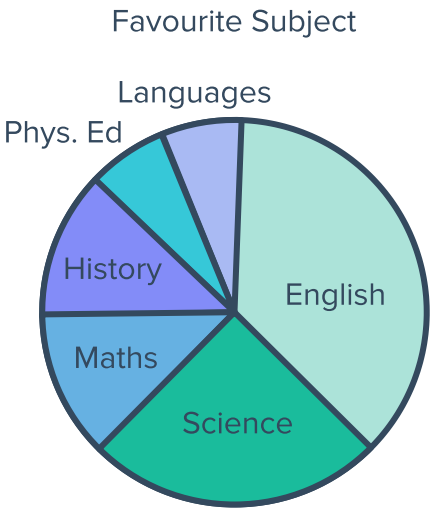
8 The results of a senatorial election in a European country, where 4000 people voted, are shown in the given pie chart:



a Who won the election?

b After the election, the winner decides to send promotional packages to everyone they suspect voted for Zorg. How many packages should they prepare?

9 Every student in year 7 was surveyed on their favourite subject, and the results are displayed in the given pie chart:



a Which was the most popular subject?

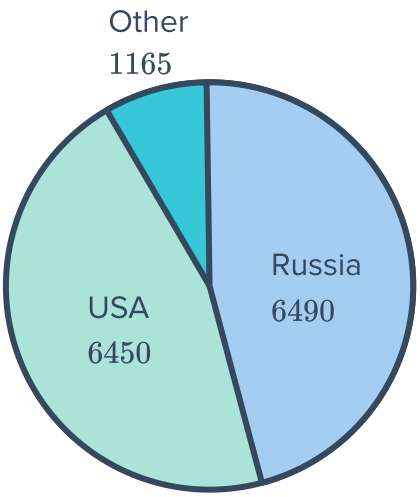
b What percentage of the class selected History, Phys. Ed., or Languages?

c If 32 students selected Science, how many students are there in year 7?

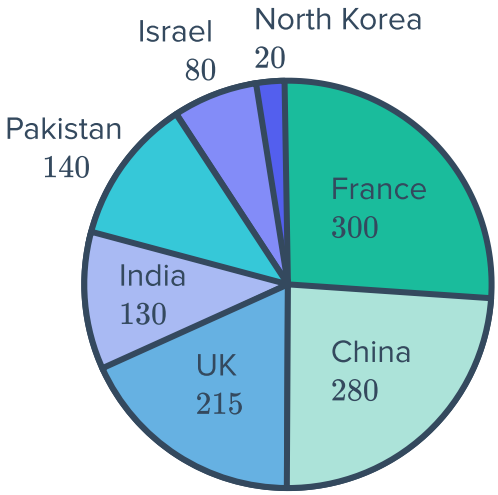
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10 The following pie charts show the distribution of nuclear weapons owned by countries around the world:

Number of Nuclear Weapons



Nuclear Weapons
(Not in USA or Russia)



a Which country has the most nuclear weapons according to the pie charts?

b Which two countries have around half of the nuclear weapons that are not owned by Russia or the USA?

c How many countries have more than 100 nuclear weapons?

Construct pie charts

11 A pharmaceutical company runs trials of a potential virus vaccine. There are 1080 volunteers for the trial, and the results are shown in the following table:

Result	Number of volunteers	Angle size
Improvement	705	$\frac{705}{1080} \times 360^\circ = 235^\circ$
No change	240	
Deterioration	135	

a Use your calculator to complete the table and find the angle size for each result.

b Hence construct a pie chart to represent this data, labelling each sector piece with the name and angle size.

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12 A business wishes to construct a pie chart to display which month holidays are taken over the summer months. They surveyed 180 employees, and the results are shown in the following table:

Month	Number of employees	Angle size
November	30	$\frac{30}{180} \times 360^\circ = 60^\circ$
December	40	
January	50	
February	20	
March	40	

a Use your calculator to complete the table and find the angle size for each month.

b Hence construct a pie chart to represent this data, labelling each sector piece with the name and angle size.

13 A marketing company wishes to construct a pie chart to display the market share of smartphone manufacturers. They surveyed 4000 people, and the results are shown in the following table:

Manufacturer	Responses	Angle size
Apple	2051	$\frac{2051}{4000} \times 360^\circ = 184.6^\circ$
Samsung	967	
Huawei	531	
Other	451	

a Use your calculator to complete the table and find the angle size for each manufacturer, correct to one decimal place.

b Hence construct a pie chart to represent this data.

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14 A teacher surveys 220 students to find their favourite subject and wishes to construct a pie chart to display the data. The survey results are shown in the following table:

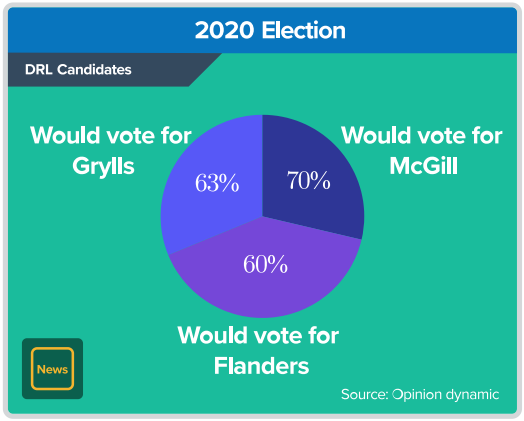
Subject	Number of students	Percentage	Angle size
Maths	44	$\frac{44}{220} = 20\%$	20% of $220 = 72^\circ$
English	55		
History	33		
Science	77		
Other	11		

a Use your calculator to complete the table and find the percentage of students and angle size for each subject.

b Hence construct a pie chart to represent this data, labelling each sector piece with the name and percentage.

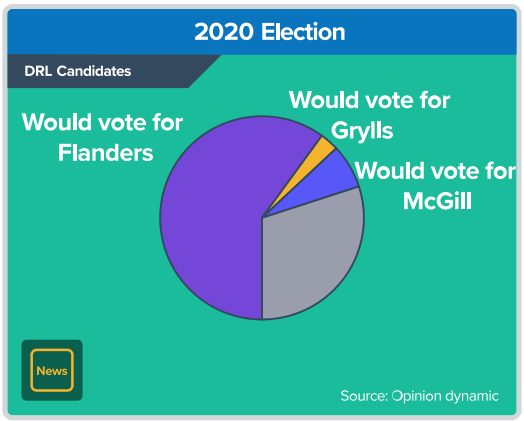
15 The given pie chart was displayed on a news program:

a Describe two mathematical errors with this pie chart.

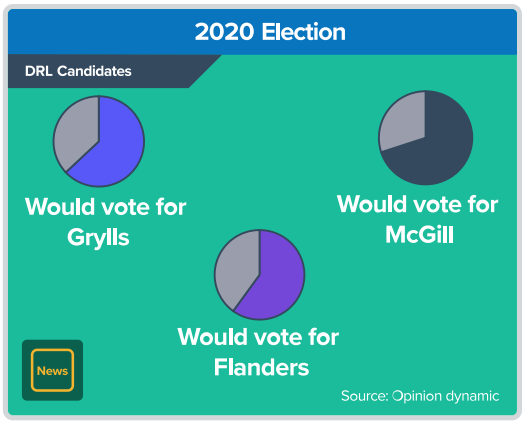


b Which of the following images better represents the same information?

A



B



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